TOPIC 4 • THEORY Advanced Mathematics • BSc Aerospace Engineering • UC3M

Systems of First-Order ODEs and the Phase Plane

From one unknown to several: how any ODE becomes a first-order system, why the theory of Topics 1 and 3 carries over, and how the eigenvalues and eigenvectors of a 2×2 matrix decide the whole geometry of the solutions — nodes, saddles, spirals, centres and stars.

Ground truth: Robinson (2004) Ch. 25 (pp. 249–252), Ch. 26 (pp. 253–258), Ch. 27 (pp. 259–268), Ch. 28 (pp. 269–284), Ch. 29 (pp. 285–293), Ch. 30 (pp. 295–300), Ch. 31 (pp. 301–305), Appendix B (pp. 382–385) • Lecture map, Part IV (Oct 6, Oct 8)

≈ 30 min study time • 29 displayed equations • 6 figures

SECTION 01 • ROBINSON CH. 25 • LECTURE OCT 6

1. Vector equations and their theory

A system of n coupled first-order equations [Robinson, Eq. (25.1), p. 249]

$$\dot{x}_i = f_i(x_1, \dots, x_n, t), \qquad i = 1, \dots, n, \tag{25.1}$$

is written compactly with $\mathbf{x} = (x_1, \dots, x_n)^T$ and $\mathbf{f} = (f_1, \dots, f_n)^T$ as

$$\frac{d\mathbf{x}}{dt} = \mathbf{f}(\mathbf{x}, t).$$

For $n = 1$ this is the scalar equation of Topic 1, and the theory is the same with vectors in place of numbers.

DEFINITION 25.1 — SOLUTION OF A VECTOR IVP [ROBINSON, P. 250]
A solution of $\dot{\mathbf{x}} = \mathbf{f}(\mathbf{x}, t), \mathbf{x}(t_0) = \mathbf{x}_0 \in \mathbb{R}^n$ (25.2) on an open interval $I \ni t_0$ is a differentiable function $\mathbf{x} : I \to \mathbb{R}^n$ with $\mathbf{x}(t_0) = \mathbf{x}_0$ and $\dot{\mathbf{x}}(t) = \mathbf{f}(\mathbf{x}(t), t)$ for all $t \in I$.

The role of $\partial f / \partial x$ is now played by the matrix of all partial derivatives, $D\mathbf{f}$ (usually called the **Jacobian**; Robinson just says "the matrix of partial derivatives") [Eq. (25.3), p. 250]:

$$D\mathbf{f} = \begin{pmatrix} \partial f_1/\partial x_1 & \cdots & \partial f_1/\partial x_n \\ \vdots & & \vdots \\ \partial f_n/\partial x_1 & \cdots & \partial f_n/\partial x_n \end{pmatrix}. \tag{25.3}$$

THEOREM 25.2 — EXISTENCE AND UNIQUENESS FOR SYSTEMS [ROBINSON, P. 250]

If $\mathbf{f}(\mathbf{x}, t)$ and $D\mathbf{f}(\mathbf{x}, t)$ are continuous for $\mathbf{x} \in U = (a_1, b_1) \times \cdots \times (a_n, b_n)$ and $c < t < d$, then for any $\mathbf{x}_0 \in U$ and $t_0 \in (c, d)$ the IVP (25.2) has a unique solution on some open interval containing t_0.

In practice: check that every f_i and every partial derivative $\partial f_i / \partial x_j$ is continuous.

EXAMPLE — THE COMPETING SPECIES OF TOPIC 1

For $\mathbf{f} = \big(x(4 - 2x - y),\ y(9 - 3x - 3y)\big)$, $D\mathbf{f} = \begin{pmatrix} 4 - 4x - y & -x \\ -3y & 9 - 3x - 6y \end{pmatrix}$. All entries are polynomials, hence continuous on all of $\mathbb{R}^2$: every initial condition has a unique local solution, and in particular two trajectories of Figure 1.1 of Topic 1 can never cross.

1.1 Higher-order equations as systems [Robinson, Sec. 25.1, pp. 251–252]

For $\ddot{x} = f(\dot{x}, x, t)$ introduce $x_1 = x$ and $x_2 = \dot{x}$. Then

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = \ddot{x} = f(x_2, x_1, t), \qquad \text{i.e.} \qquad \frac{d\mathbf{x}}{dt} = \mathbf{g}(\mathbf{x}, t) = \begin{pmatrix} x_2 \\ f(x_2, x_1, t) \end{pmatrix}. \tag{25.4}$$

The vector initial condition $\mathbf{x}(t_0) = \mathbf{x}_0$ is exactly the pair $x(t_0), \dot{x}(t_0)$, and $D\mathbf{g} = \begin{pmatrix} 0 & 1 \\ \partial f/\partial x_1 & \partial f/\partial x_2 \end{pmatrix}$. Applying Theorem 25.2 gives **Theorem 25.3**, which is Theorem 11.1 of Topic 3. An nth-order equation becomes an n-dimensional system in the same way, with $x_k = x^{(k-1)}$.

EXAMPLE — THE MASS-SPRING-DAMPER AS A SYSTEM

$\ddot{x} + \lambda \dot{x} + \omega^2 x = 0$ with $x_1 = x$ (position) and $x_2 = \dot{x}$ (velocity) becomes

$$\frac{d}{dt}\begin{pmatrix} x_1 \\ x_2 \end{pmatrix} = \begin{pmatrix} 0 & 1 \\ -\omega^2 & -\lambda \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}.$$

The plane of $(x, \dot{x})$ is the **phase plane**; we return to this example in Section 9.

SECTION 02 • ROBINSON CH. 26 • LECTURE OCT 6

2. Linear systems and the elimination method

A first-order linear system of two equations with constant coefficients is [Robinson, Eq. (26.1), p. 253]

$$\dot{x} = ax + by + f(t), \qquad \dot{y} = cx + dy + g(t). \tag{26.1}$$

It is **homogeneous** if $f = g = 0$ and **inhomogeneous** otherwise. If $b = c = 0$ the equations are uncoupled and each is solved by an integrating factor.

2.1 Reduction to one second-order equation

Assume $b \neq 0$ (otherwise exchange the roles of x and y).

1. Solve the first equation for y: $y = \dfrac{\dot x - a x - f(t)}{b}$ (26.2).
2. Differentiate: $\dot y = \dfrac{\ddot x - a\dot x - \dot f(t)}{b}$.
3. Substitute both into the second equation: $\dfrac{\ddot x - a\dot x - \dot f}{b} = c x + d\,\dfrac{\dot x - a x - f}{b} + g$.
4. Multiply by b and collect terms:

$$\ddot x - (a + d)\,\dot x + (ad - bc)\,x = \dot f(t) + b\,g(t) - d\,f(t). \tag{26.3}$$

Solve (26.3) by the methods of Topic 3, then recover y from (26.2). The coefficients $a + d$ and $ad - bc$ are the trace and the determinant of the matrix $\begin{pmatrix} a & b \\ c & d \end{pmatrix}$ – they will reappear in the eigenvalue method.

EXAMPLE 26.1 (PP. 254–255): $\dot x = x + y$, $\dot y = 4x - 2y + 4e^{-2t}$, $x(0) = 0$, $y(0) = -1$

1. Here $a = 1, b = 1, c = 4, d = -2$, $f = 0, g = 4e^{-2t}$. From the first equation $y = \dot x - x$ (26.4), so $\dot y = \ddot x - \dot x$.
2. Substituting: $\ddot x - \dot x = 4x - 2(\dot x - x) + 4e^{-2t}$, i.e. $\ddot x + \dot x - 6x = 4e^{-2t}$ (26.5) – as (26.3) predicts: $a + d = -1$, $ad - bc = -6$, right side $b\,g = 4e^{-2t}$.
3. Complementary function: $k^2 + k - 6 = 0$, $k = 2, -3$. Particular integral Ce^{-2t}: $(4 - 2 - 6)C = 4$, $C = -1$. So $x = Ae^{2t} + Be^{-3t} - e^{-2t}$.
4. $y = \dot x - x = (2A - A)e^{2t} + (-3B - B)e^{-3t} + (2 + 1)e^{-2t} = Ae^{2t} - 4Be^{-3t} + 3e^{-2t}$.
5. Initial data: $A + B - 1 = 0$ and $A - 4B + 3 = -1$. Subtracting, $5B = 5$: $B = 1$, $A = 0$.

$$x(t) = e^{-3t} - e^{-2t}, \qquad y(t) = 3e^{-2t} - 4e^{-3t}.$$

Check: $\dot x = -3e^{-3t} + 2e^{-2t}$ and $x + y = 2e^{-2t} - 3e^{-3t}$; $x(0) = 0, y(0) = -1$.

EXAMPLE 26.2 (PP. 255–256): $\dot x = 2x + 5y$, $\dot y = -2x$, $x(0) = 5$, $y(0) = -4$

$y = \tfrac{1}{5}(\dot x - 2x)$, so $\tfrac{1}{5}(\ddot x - 2\dot x) = -2x$, i.e. $\ddot x - 2\dot x + 10x = 0$, with roots $k = 1 \pm 3i$: $x = e^t(A\cos 3t + B\sin 3t)$ and $y = \tfrac{e^t}{5}\big[(3B - A)\cos 3t - (3A + B)\sin 3t\big]$. The data give $A = 5$, $B = -5$: $x = 5e^t(\cos 3t - \sin 3t)$, $y = -e^t(4\cos 3t + 2\sin 3t)$ – oscillations with exponentially growing amplitude.

3. Matrix form, superposition and the fundamental matrix

The homogeneous system $\dot{x} = ax + by, \dot{y} = cx + dy$ (27.1) is, with $\mathbf{x} = (x, y)^T$ [Eq. (27.2), p. 259],

$$\dot{\mathbf{x}} = A\mathbf{x}, \qquad A = \begin{pmatrix} a & b \\ c & d \end{pmatrix}. \tag{27.2}$$

SUPERPOSITION [ROBINSON, EQ. (27.3), P. 260]
If $\mathbf{x}_1$ and $\mathbf{x}_2$ solve (27.2), so does $\mathbf{x} = \alpha\mathbf{x}_1 + \beta\mathbf{x}_2$: $\dot{\mathbf{x}} = \alpha\dot{\mathbf{x}}_1 + \beta\dot{\mathbf{x}}_2 = \alpha A\mathbf{x}_1 + \beta A\mathbf{x}_2 = A(\alpha\mathbf{x}_1 + \beta\mathbf{x}_2) = A\mathbf{x}$.

LECTURE MATERIAL NOT IN ROBINSON (LECTURE MAP, OCT 6)
The lecture developed the general theory of linear systems – linear independence of vector solutions, the Wronskian, fundamental sets and the "homogeneous + particular" structure – which Robinson does not treat for systems. The nearest book material is the scalar version: Sec. 11.3–11.4 (independence, Wronskian) and Sec. 14.1 (homogeneous + particular). The statements below follow the lecture; the proofs mirror Topic 3.

DEFINITIONS – INDEPENDENCE, WRONSKIAN, FUNDAMENTAL MATRIX

- Vector functions $\mathbf{x}_1(t), \mathbf{x}_2(t)$ are **linearly independent** on I if $\alpha\mathbf{x}_1(t) + \beta\mathbf{x}_2(t) = \mathbf{0}$ for all $t \in I$ forces $\alpha = \beta = 0$.
- Their **Wronskian** is the determinant of the matrix whose columns are the solutions: $W(t) = \det\begin{bmatrix}\mathbf{x}_1(t) & \mathbf{x}_2(t)\end{bmatrix} = x_{11}x_{22} - x_{12}x_{21}$, where $\mathbf{x}_j = (x_{1j}, x_{2j})^T$.
- Two independent solutions form a **fundamental set**; the matrix $\Phi(t) = [\mathbf{x}_1 \;\; \mathbf{x}_2]$ is a **fundamental matrix**. It satisfies $\dot{\Phi} = A\Phi$, column by column.

THEOREM – WRONSKIAN TEST AND GENERAL SOLUTION FOR SYSTEMS

1. Two solutions of $\dot{\mathbf{x}} = A\mathbf{x}$ are linearly independent if and only if $W(t) \neq 0$; and then $W(t) \neq 0$ for all t.
2. If Φ is a fundamental matrix, every solution is $\mathbf{x}(t) = \Phi(t)\,\mathbf{c} = c_1\mathbf{x}_1(t) + c_2\mathbf{x}_2(t)$, and the IVP $\mathbf{x}(t_0) = \mathbf{x}_0$ has $\mathbf{c} = \Phi(t_0)^{-1}\mathbf{x}_0$.

PROOF
If $W(t_0) \neq 0$, the columns $\mathbf{x}_1(t_0), \mathbf{x}_2(t_0)$ are independent vectors, so $\alpha\mathbf{x}_1(t_0) + \beta\mathbf{x}_2(t_0) = \mathbf{0}$ already forces $\alpha = \beta = 0$. Conversely, if $W(t_0) = 0$, there are $(\alpha, \beta) \neq (0, 0)$ with $\alpha\mathbf{x}_1(t_0) + \beta\mathbf{x}_2(t_0) = \mathbf{0}$. Then $\mathbf{z} = \alpha\mathbf{x}_1 + \beta\mathbf{x}_2$ solves the system (superposition) with $\mathbf{z}(t_0) = \mathbf{0}$; so does $\mathbf{0}$, and by uniqueness (Theorem 25.2) $\mathbf{z} \equiv \mathbf{0}$: the solutions are dependent. For (2): given any solution $\mathbf{x}$, the system $\Phi(t_0)\mathbf{c} = \mathbf{x}(t_0)$ is solvable because $\det\Phi(t_0) = W(t_0) \neq 0$; then $\Phi\mathbf{c}$ and $\mathbf{x}$ are solutions with the same value at t_0, hence equal for all t.

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Liouville's formula – Abel's identity for systems (complementary)

Write the columns as $\mathbf{x}_j = (p_j, q_j)^T$, so $\dot p_j = a p_j + b q_j$, $\dot q_j = c p_j + d q_j$ and $W = p_1 q_2 - q_1 p_2$. By the product rule

$$\dot W = \dot p_1 q_2 + p_1 \dot q_2 - \dot q_1 p_2 - q_1 \dot p_2 = (a p_1 + b q_1) q_2 + p_1 (c p_2 + d q_2) - (c p_1 + d q_1) p_2 - q_1 (a p_2 + b q_2).$$

The b terms ($b q_1 q_2 - b q_1 q_2$) and the c terms ($c p_1 p_2 - c p_1 p_2$) cancel, leaving

$$\dot W = (a + d)(p_1 q_2 - q_1 p_2) = (\operatorname{tr} A)\, W \quad \Longrightarrow \quad W(t) = W(t_0)\, e^{(\operatorname{tr} A)(t - t_0)}.$$

(The computation never uses that a, b, c, d are constant; with time-dependent entries $W(t) = W(t_0) \exp \int_{t_0}^{t} \operatorname{tr} A(s)\, ds$.) For a second-order equation turned into a system, $A = \begin{pmatrix} 0 & 1 \\ -q & -p \end{pmatrix}$ has trace $-p$, the solution columns are $(x_j, \dot x_j)^T$, and W is exactly the scalar Wronskian $x_1 \dot x_2 - x_2 \dot x_1$: Abel's identity of Topic 3 is the special case.

3.1 Inhomogeneous systems

LEMMA – GENERAL SOLUTION OF $\dot{\mathbf{x}} = A\mathbf{x} + \mathbf{f}(t)$ (LECTURE; SCALAR ANALOGUE ROBINSON SEC. 14.1)

If $\mathbf{x}_p$ is one solution, every solution is $\mathbf{x} = \Phi(t)\mathbf{c} + \mathbf{x}_p(t)$.

PROOF

If $\mathbf{x}$ solves the inhomogeneous system, then $\frac{d}{dt}(\mathbf{x} - \mathbf{x}_p) = (A\mathbf{x} + \mathbf{f}) - (A\mathbf{x}_p + \mathbf{f}) = A(\mathbf{x} - \mathbf{x}_p)$, so $\mathbf{x} - \mathbf{x}_p$ solves the homogeneous system and equals $\Phi\mathbf{c}$. Conversely $\frac{d}{dt}(\Phi\mathbf{c} + \mathbf{x}_p) = A\Phi\mathbf{c} + A\mathbf{x}_p + \mathbf{f}$.

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Variation of parameters for systems (complementary). As in Topic 3, let the constants vary: $\mathbf{x}_p = \Phi(t)\mathbf{c}(t)$. The product rule gives $\dot{\mathbf{x}}_p = \dot\Phi\mathbf{c} + \Phi\dot{\mathbf{c}} = A\Phi\mathbf{c} + \Phi\dot{\mathbf{c}}$; this equals $A\mathbf{x}_p + \mathbf{f}$ exactly when $\Phi\dot{\mathbf{c}} = \mathbf{f}$. Since $\det\Phi = W \neq 0$:

$$\mathbf{x}_p(t) = \Phi(t) \int \Phi(s)^{-1}\, \mathbf{f}(s)\, ds.$$

EXAMPLE 26.1 AGAIN, BY A FUNDAMENTAL MATRIX

The homogeneous solutions are $e^{2t}(1,1)^T$ and $e^{-3t}(1,-4)^T$ (Section 5), so $\Phi = \begin{pmatrix} e^{2t} & e^{-3t} \\ e^{2t} & -4e^{-3t} \end{pmatrix}$, $W = -4e^{-t} - e^{-t} = -5e^{-t}$ (check: $\operatorname{tr} A = -1$, so $W \propto e^{-t}$), and $\Phi^{-1} = \frac{1}{W}\begin{pmatrix} -4e^{-3t} & -e^{-3t} \\ -e^{2t} & e^{2t} \end{pmatrix} = \frac{1}{5}\begin{pmatrix} 4e^{-2t} & e^{-2t} \\ e^{3t} & -e^{3t} \end{pmatrix}$. With $\mathbf{f} = (0, 4e^{-2t})^T$: $\Phi^{-1}\mathbf{f} = \frac{4}{5}\left(e^{-4t},\ -e^{t}\right)^T$, whose antiderivative is $\left(-\frac{1}{5}e^{-4t},\ -\frac{4}{5}e^{t}\right)^T$. Then

$$\mathbf{x}_p = \Phi\begin{pmatrix} -\frac{1}{5}e^{-4t} \\ -\frac{4}{5}e^{t} \end{pmatrix} = \begin{pmatrix} -\frac{1}{5}e^{-2t} - \frac{4}{5}e^{-2t} \\ -\frac{1}{5}e^{-2t} + \frac{16}{5}e^{-2t} \end{pmatrix} = \begin{pmatrix} -e^{-2t} \\ 3e^{-2t} \end{pmatrix},$$

the same particular solution that elimination produced .

SECTION 04 • ROBINSON SEC. 27.1-27.2 • LECTURE OCT 6-8

4. The eigenvalue method

For the scalar equation $\dot{x} = ax$ the solutions are $e^{at}x_0$. By analogy try a solution that keeps a fixed direction $\mathbf{v}$ and changes only in size [Eq. (27.4), p. 260]:

$$\mathbf{x}(t) = e^{\lambda t}\mathbf{v}, \qquad \mathbf{v} \text{ constant},\ \mathbf{v} \neq \mathbf{0}. \tag{27.4}$$

Then $\dot{\mathbf{x}} = \lambda e^{\lambda t}\mathbf{v}$ and $A\mathbf{x} = e^{\lambda t}A\mathbf{v}$; equating and dividing by $e^{\lambda t} \neq 0$:

$$A\mathbf{v} = \lambda\mathbf{v}. \tag{27.5}$$

So $e^{\lambda t}\mathbf{v}$ is a solution exactly when λ is an **eigenvalue** of A and $\mathbf{v}$ a corresponding **eigenvector**.

4.1 Finding eigenvalues and eigenvectors [Sec. 27.2, pp. 260-261]

1. Rewrite (27.5) as $(A-\lambda I)\mathbf{v}=\mathbf{0}$ (27.7). If $A-\lambda I$ were invertible, the only solution would be $\mathbf{v}=\mathbf{0}$. A non-zero $\mathbf{v}$ therefore requires $\det(A-\lambda I)=0$.

2. Expand: $\det\begin{pmatrix} a-\lambda & b \\ c & d-\lambda \end{pmatrix} = (a-\lambda)(d-\lambda)-bc$, giving the **characteristic equation**

$$\lambda^2-(a+d)\lambda+(ad-bc)=0, \qquad \text{i.e.} \qquad \lambda^2-(\operatorname{tr}A)\lambda+\det A=0. \quad (27.8)$$

3. For each root λ, solve the (singular) system $(A-\lambda I)\mathbf{v}=\mathbf{0}$: one row is a multiple of the other, so one equation fixes the direction of $\mathbf{v}$. Any non-zero multiple is also an eigenvector, $A(c\mathbf{v})=c\lambda\mathbf{v}$.

Equation (27.8) is the characteristic equation (26.3) of the eliminated second-order equation: both methods see the same numbers. Its roots satisfy $\lambda_1+\lambda_2=\operatorname{tr}A$ and $\lambda_1\lambda_2=\det A$.

EXAMPLE 27.1 (PP. 262-263): $A=\begin{pmatrix} 2 & 2 \\ 6 & 3 \end{pmatrix}$

1. $(2-\lambda)(3-\lambda)-12=\lambda^2-5\lambda-6=(\lambda-6)(\lambda+1)=0$: $\lambda_1=6$, $\lambda_2=-1$. (Check: sum $5=\operatorname{tr}A$, product $-6=\det A$.)

2. $\lambda=6$: $(A-6I)\mathbf{v}=\begin{pmatrix} -4 & 2 \\ 6 & -3 \end{pmatrix}\mathbf{v}=\mathbf{0}$ gives $-4v_1+2v_2=0$, $v_2=2v_1$: $\mathbf{v}_1=(1,2)^T$. Check: $A\mathbf{v}_1=(6,12)^T=6\mathbf{v}_1$.

3. $\lambda=-1$: $\begin{pmatrix} 3 & 2 \\ 6 & 4 \end{pmatrix}\mathbf{v}=\mathbf{0}$ gives $3v_1+2v_2=0$: $\mathbf{v}_2=(2,-3)^T$. Check: $A\mathbf{v}_2=(-2,3)^T=-\mathbf{v}_2$.

TRIANGULAR MATRICES (EXAMPLES 27.2-27.3, P. 264)

For $\begin{pmatrix} \lambda_1 & b \\ 0 & \lambda_2 \end{pmatrix}$ the characteristic polynomial is $(\lambda_1-\lambda)(\lambda_2-\lambda)-b\cdot 0$, so the eigenvalues are the diagonal entries, and $(1,0)^T$ is an eigenvector for λ_1. The same holds for lower-triangular matrices (with $(0,1)^T$ for λ_2) and, with both eigenvectors along the axes, for diagonal matrices. Robinson simply treats matrices of the form $\begin{pmatrix} \lambda_1 & b \\ 0 & \lambda_2 \end{pmatrix}$ without naming them; the lecture calls them triangular.

The roots of (27.8) fall into three cases, each with its own chapter [p. 261]:

DISCRIMINANT $(\operatorname{tr}A)^2-4\det A$	EIGENVALUES	EXAMPLE	TREATED IN
>0	Real, distinct	27.1: $6,-1$	Ch. 28 – Sections 5–6
$=0$	One repeated real	27.4: $A=\begin{pmatrix} 5 & -4 \\ 1 & 1 \end{pmatrix}$, $(\lambda-3)^2=0$, $\mathbf{v}=(2,1)^T$ only	Ch. 30 – Section 8
<0	Complex pair $\rho\pm i\omega$	27.5: $A=\begin{pmatrix} 2 & 5 \\ -2 & 0 \end{pmatrix}$, $\lambda=1\pm 3i$	Ch. 29 – Section 7

5. Distinct real eigenvalues: solution and change of basis

If $\lambda_1 \neq \lambda_2$ are real with eigenvectors $\mathbf{v}_1, \mathbf{v}_2$, then $e^{\lambda_1 t}\mathbf{v}_1$ and $e^{\lambda_2 t}\mathbf{v}_2$ are solutions, and by superposition [Eq. (28.1), p. 270]

$$\mathbf{x}(t) = Ae^{\lambda_1 t}\mathbf{v}_1 + Be^{\lambda_2 t}\mathbf{v}_2. \tag{28.1}$$

This is the general solution because the two solutions are independent. Their Wronskian at $t = 0$ is $\det[\mathbf{v}_1\ \mathbf{v}_2]$, non-zero by the next result.

PROPOSITION — EIGENVECTORS OF DISTINCT EIGENVALUES ARE INDEPENDENT [ROBINSON, APPENDIX B, P. 384]
If $A\mathbf{v}_1 = \lambda_1\mathbf{v}_1$, $A\mathbf{v}_2 = \lambda_2\mathbf{v}_2$ with $\mathbf{v}_1, \mathbf{v}_2 \neq \mathbf{0}$ and $\lambda_1 \neq \lambda_2$, then $\mathbf{v}_1, \mathbf{v}_2$ are linearly independent.

PROOF
Suppose $\alpha\mathbf{v}_1 + \beta\mathbf{v}_2 = \mathbf{0}$ (B.3). Applying A gives $\alpha\lambda_1\mathbf{v}_1 + \beta\lambda_2\mathbf{v}_2 = \mathbf{0}$ (B.4). Multiply (B.3) by λ_1 and subtract (B.4):

$$\beta(\lambda_1 - \lambda_2)\mathbf{v}_2 = \mathbf{0}.$$

Since $\lambda_1 \neq \lambda_2$ and $\mathbf{v}_2 \neq \mathbf{0}$, $\beta = 0$; then (B.3) reads $\alpha\mathbf{v}_1 = \mathbf{0}$, so $\alpha = 0$. (Robinson argues by dividing by α and λ_1; subtracting as here avoids having to treat $\alpha = 0$ or $\lambda_1 = 0$ separately.)

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EXAMPLE 28.1 (PP. 270–271): $\dot{x} = x + y,\ \dot{y} = 4x - 2y$
$A = \begin{pmatrix} 1 & 1 \\ 4 & -2 \end{pmatrix}$: $(1-\lambda)(-2-\lambda) - 4 = \lambda^2 + \lambda - 6 = 0, \lambda = 2, -3$. $\lambda = 2$: $(A - 2I) = \begin{pmatrix} -1 & 1 \\ 4 & -4 \end{pmatrix}$, $\mathbf{v}_1 = (1,1)^T$. $\lambda = -3$: $(A + 3I) = \begin{pmatrix} 4 & 1 \\ 4 & 1 \end{pmatrix}$, $\mathbf{v}_2 = (1,-4)^T$. Independence directly: $\alpha(1,1) + \beta(1,-4) = \mathbf{0}$ gives $\alpha + \beta = 0, \alpha - 4\beta = 0$, so $\alpha = \beta = 0$.

$$\mathbf{x}(t) = Ae^{2t}\begin{pmatrix} 1 \\ 1 \end{pmatrix} + Be^{-3t}\begin{pmatrix} 1 \\ -4 \end{pmatrix}.$$

This is the homogeneous part of Example 26.1, as it should be.

5.1 Changing coordinates: diagonalisation [Sec. 28.2, pp. 271–274]

Because $\mathbf{v}_1, \mathbf{v}_2$ are independent, every vector can be written in the eigenvector basis, $\mathbf{x} = \tilde{x}\mathbf{v}_1 + \tilde{y}\mathbf{v}_2$ (28.3). In matrix form, with $P = [\mathbf{v}_1 \ \mathbf{v}_2]$ (eigenvectors as columns),

$$\mathbf{x} = P\tilde{\mathbf{x}}, \qquad \tilde{\mathbf{x}} = P^{-1}\mathbf{x}. \tag{28.4–28.5}$$

(The lecture writes $\mathbf{z}$ for Robinson's $\tilde{\mathbf{x}}$.) Differentiate the second relation; P^{-1} is constant:

$$\frac{d\tilde{\mathbf{x}}}{dt} = P^{-1}\frac{d\mathbf{x}}{dt} = P^{-1}A\mathbf{x} = P^{-1}AP\,\tilde{\mathbf{x}}. \tag{28.7}$$

Now compute $P^{-1}AP$ column by column, using $A[\mathbf{v}_1 \ \mathbf{v}_2] = [A\mathbf{v}_1 \ A\mathbf{v}_2]$:

$$AP = [\lambda_1\mathbf{v}_1 \ \ \lambda_2\mathbf{v}_2] = [\mathbf{v}_1 \ \mathbf{v}_2]\begin{pmatrix}\lambda_1 & 0\\ 0 & \lambda_2\end{pmatrix} = P\Lambda \quad\Longrightarrow\quad P^{-1}AP = \Lambda = \begin{pmatrix}\lambda_1 & 0\\ 0 & \lambda_2\end{pmatrix}.$$

So in the new coordinates the system is **uncoupled** [Eqs. (28.8)–(28.9)]:

$$\dot{\tilde{x}} = \lambda_1\tilde{x}, \quad \dot{\tilde{y}} = \lambda_2\tilde{y} \quad\Longrightarrow\quad \tilde{x} = Ae^{\lambda_1 t}, \quad \tilde{y} = Be^{\lambda_2 t},$$

and $\mathbf{x} = P\tilde{\mathbf{x}} = Ae^{\lambda_1 t}\mathbf{v}_1 + Be^{\lambda_2 t}\mathbf{v}_2$ recovers (28.1).

EXAMPLE 28.2 (PP. 274–276): DECOUPLING $\dot{x} = x + y,\ \dot{y} = 4x - 2y$

$P = \begin{pmatrix}1 & 1\\ 1 & -4\end{pmatrix}$, $\det P = -4 - 1 = -5 \neq 0$, and with $\begin{pmatrix}a & b\\ c & d\end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix}d & -b\\ -c & a\end{pmatrix}$ (Appendix B), $P^{-1} = \frac{1}{5}\begin{pmatrix}4 & 1\\ 1 & -1\end{pmatrix}$, i.e. $\tilde{x} = \frac{1}{5}(4x + y)$, $\tilde{y} = \frac{1}{5}(x - y)$. Then

$$P^{-1}AP = \frac{1}{5}\begin{pmatrix}4 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}1 & 1\\ 4 & -2\end{pmatrix}\begin{pmatrix}1 & 1\\ 1 & -4\end{pmatrix} = \frac{1}{5}\begin{pmatrix}4 & 1\\ 1 & -1\end{pmatrix}\begin{pmatrix}2 & -3\\ 2 & 12\end{pmatrix} = \frac{1}{5}\begin{pmatrix}10 & 0\\ 0 & -15\end{pmatrix} = \begin{pmatrix}2 & 0\\ 0 & -3\end{pmatrix}\ \checkmark.$$

Robinson stresses that this is not the efficient way to *solve* the system – (28.1) is – but it is the key to *drawing* it.

SECTION 06 • ROBINSON SEC. 28.3–28.5 • LECTURE OCT 8

6. Phase portraits: nodes and saddles

A **phase portrait** shows the curves traced in the (x, y) plane by solutions, with arrows for increasing t. The origin is always a stationary solution of $\dot{\mathbf{x}} = A\mathbf{x}$ (and the only one when $\det A \neq 0$). Draw first the uncoupled system, then "twist" it.

6.1 The uncoupled system `[Sec. 28.3, pp. 276–278]`

Solutions of $\dot{\tilde{x}} = \lambda_1 \tilde{x}$, $\dot{\tilde{y}} = \lambda_2 \tilde{y}$ are $\tilde{x} = \tilde{x}_0 e^{\lambda_1 t}$, $\tilde{y} = \tilde{y}_0 e^{\lambda_2 t}$ (28.12). To find the curve they trace, eliminate t: $e^t = (\tilde{x}/\tilde{x}_0)^{1/\lambda_1}$, so

$$\tilde{y} = \tilde{y}_0\, e^{\lambda_2 t} = \tilde{y}_0 \left(\frac{\tilde{x}}{\tilde{x}_0}\right)^{\lambda_2/\lambda_1} = K\, \tilde{x}^{\lambda_2/\lambda_1}, \qquad K = \tilde{y}_0\, \tilde{x}_0^{-\lambda_2/\lambda_1}.$$

(Equivalently, by the chain rule $\dfrac{d\tilde{y}}{d\tilde{x}} = \dfrac{\dot{\tilde{y}}}{\dot{\tilde{x}}} = \dfrac{\lambda_2 \tilde{y}}{\lambda_1 \tilde{x}}$, a separable equation.) The axes themselves are trajectories: data on an axis stay on it.

- **Stable node** ($\lambda_1 < \lambda_2 < 0$): every solution tends to $\mathbf{0}$. Since $|\lambda_2| < |\lambda_1|$, the $\tilde{y}$ component decays more slowly, so trajectories arrive tangent to the slow eigendirection.
- **Unstable node** ($0 < \lambda_1 < \lambda_2$): the same picture with arrows reversed; solutions leave the origin tangent to the eigendirection of smaller $|\lambda|$.
- **Saddle** ($\lambda_2 < 0 < \lambda_1$): $\tilde{x}$ grows while $\tilde{y}$ decays; the curves $\tilde{x}\,\tilde{y}^{\,r} = K$ ($r = -\lambda_1/\lambda_2 > 0$) are hyperbola-like. Only solutions starting on the $\tilde{y}$-axis reach the origin.

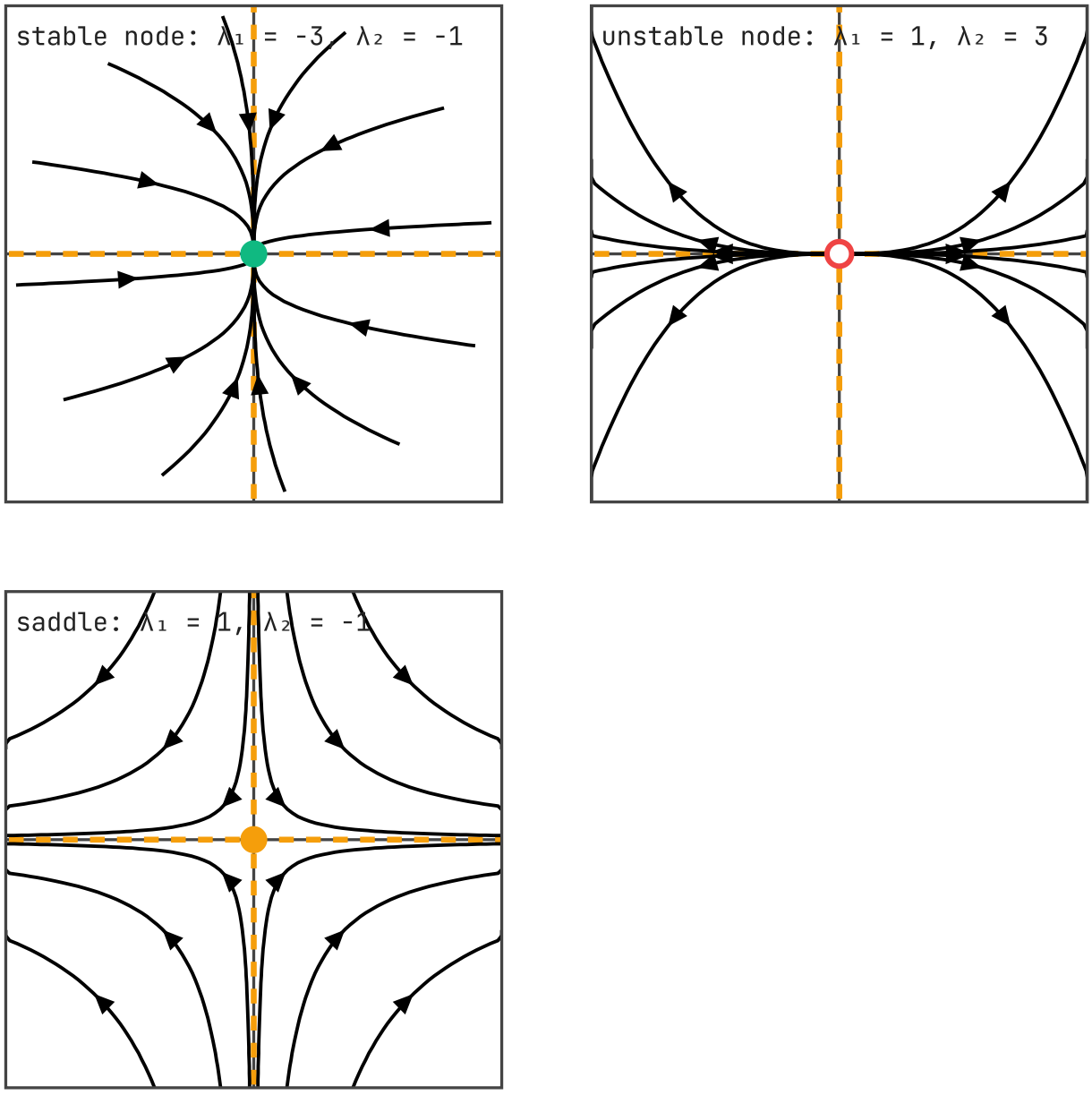


Figure 4.1. *Uncoupled systems $\dot{\tilde{x}} = \lambda_1 \tilde{x}$, $\dot{\tilde{y}} = \lambda_2 \tilde{y}$ (Robinson Sec. 28.3, Figs. 28.2–28.5). Trajectories are the curves $\tilde{y} = K\tilde{x}^{\lambda_2/\lambda_1}$. In the nodes, solutions are tangent at the origin to the axis of the eigenvalue with smaller $|\lambda|$.*

6.2 Back to the original coordinates `[Sec. 28.4, p. 279]`

The map $\mathbf{x} = P\tilde{\mathbf{x}}$ sends the $\tilde{x}$-axis to the line through $\mathbf{v}_1$ and the $\tilde{y}$-axis to the line through $\mathbf{v}_2$. Straight lines stay straight and the origin stays fixed, so the picture is the uncoupled one, sheared and rotated. Practical recipe:

1. Find the eigenvalues and eigenvectors.
2. Draw the eigenvector lines; put arrows pointing away from the origin if $\lambda > 0$ and towards it if $\lambda < 0$.
3. Fill in the other trajectories consistently with the type (node or saddle), tangent to the slow direction near the origin and parallel to the fast one far away.

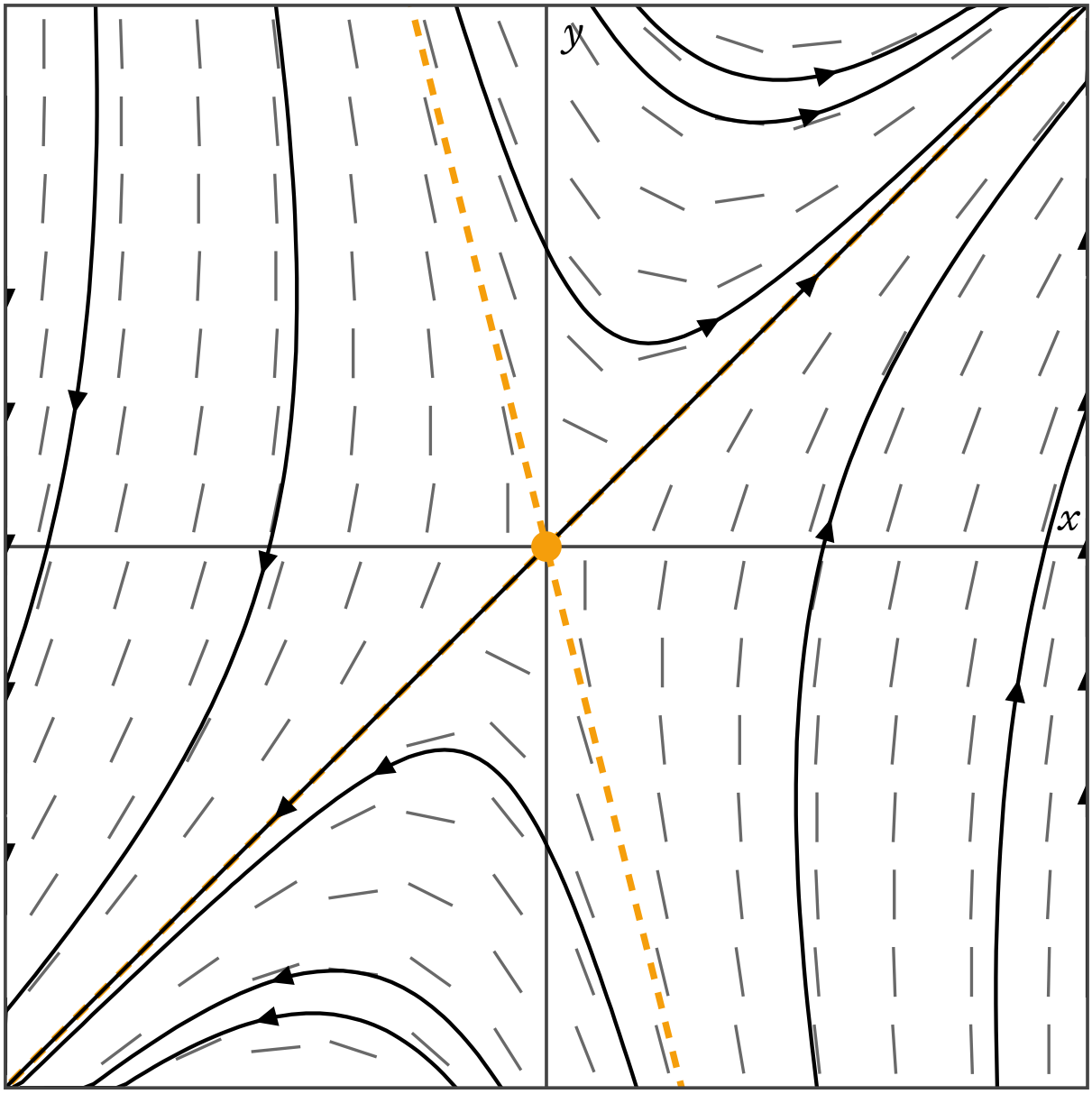


Figure 4.2. *Phase portrait of $\dot{x} = x + y$, $\dot{y} = 4x - 2y$ (Robinson Example 28.3, Fig. 28.8). Dashed amber: the eigenvector lines $y = x$ ($\lambda_1 = 2$, outward) and $y = -4x$ ($\lambda_2 = -3$, inward). The origin is a saddle; the line $y = -4x$ is its stable manifold and $y = x$ its unstable manifold.*

6.3 Stable and unstable manifolds [Sec. 28.5, pp. 281–282]

DEFINITIONS

The **stable manifold** $W^s(\mathbf{0})$ is the set of initial points whose trajectories tend to the origin as $t \to +\infty$; the **unstable manifold** $W^u(\mathbf{0})$ is the set whose trajectories tend to the origin as $t \to -\infty$.

	$W^s(\mathbf{0})$	$W^u(\mathbf{0})$
Stable node	the whole plane	$\{\mathbf{0}\}$
Saddle	the eigenvector line of the negative eigenvalue	the eigenvector line of the positive eigenvalue
Unstable node	$\{\mathbf{0}\}$	the whole plane

At a saddle the stable manifold is a **separatrix**: points on one side are carried to infinity along one branch of W^u, points on the other side along the opposite branch. In Example 28.3, initial data above the line $y = -4x$ end up with $x, y \to +\infty$; data below it with $x, y \to -\infty$.

TERMINOLOGY

The lecture notes mention a "saddle node"; in the phase-portrait chapters Robinson says only "saddle point" or "saddle" and reserves "node" for stable and unstable nodes, so the intended term is almost certainly "saddle". (In bifurcation theory, Topic 2, a *saddle-node* is something else.)

SECTION 07 • ROBINSON CH. 29 (BEYOND THE OCT 8 LECTURE)

7. Complex eigenvalues: spirals and centres

Let $\lambda_\pm = \rho \pm i\omega$ ($\omega > 0$) with eigenvectors $\boldsymbol{\eta}_\pm = \mathbf{v}_1 \pm i\mathbf{v}_2$, where $\mathbf{v}_1, \mathbf{v}_2$ are real (for a real matrix the eigenvector of $\bar\lambda$ is the conjugate of that of λ).

7.1 Real solutions [Sec. 29.1, pp. 285–286]

1. For a real solution take conjugate coefficients: $\mathbf{x} = Ce^{\lambda_+ t}\boldsymbol{\eta}_+ + C^* e^{\lambda_- t}\boldsymbol{\eta}_- = 2\,\mathrm{Re}\big[Ce^{(\rho+i\omega)t}(\mathbf{v}_1 + i\mathbf{v}_2)\big]$ (29.1), with $C = \alpha + i\beta$.

2. Euler: $Ce^{i\omega t} = (\alpha\cos\omega t - \beta\sin\omega t) + i(\beta\cos\omega t + \alpha\sin\omega t)$. Multiply by $\mathbf{v}_1 + i\mathbf{v}_2$ and keep the real part: $(\alpha\cos\omega t - \beta\sin\omega t)\mathbf{v}_1 - (\beta\cos\omega t + \alpha\sin\omega t)\mathbf{v}_2$.

3. With $A = 2\alpha$, $B = -2\beta$:

$$\mathbf{x}(t) = e^{\rho t}\Big[\big(A\cos\omega t + B\sin\omega t\big)\mathbf{v}_1 + \big(B\cos\omega t - A\sin\omega t\big)\mathbf{v}_2\Big]. \tag{29.2}$$

EXAMPLE 29.1 (P. 286): $\dot{x} = 2x + 5y,\ \dot{y} = -2x$

From Example 27.5, $\lambda = 1 \pm 3i$ and, for $\lambda = 1 + 3i$, the first row $(1 - 3i)v_1 + 5v_2 = 0$ gives $\boldsymbol{\eta} = (5,\ -1 + 3i)^T = (5, -1)^T + i(0, 3)^T$. (Check: $A\boldsymbol{\eta} = (10 - 5 + 15i,\ -10)^T = (5 + 15i,\ -10)^T$ and $(1 + 3i)\boldsymbol{\eta} = (5 + 15i,\ -1 + 3i - 3i - 9)^T$.) By (29.2), $x = 5e^t(A\cos 3t + B\sin 3t)$, $y = e^t\big[(3B - A)\cos 3t - (3A + B)\sin 3t\big]$, the same as Example 26.2.

7.2 Canonical form and polar coordinates [Sec. 29.2, pp. 287–290]

Taking real and imaginary parts of $A(\mathbf{v}_1 + i\mathbf{v}_2) = (\rho + i\omega)(\mathbf{v}_1 + i\mathbf{v}_2)$:

$$A\mathbf{v}_1 = \rho\mathbf{v}_1 - \omega\mathbf{v}_2, \qquad A\mathbf{v}_2 = \omega\mathbf{v}_1 + \rho\mathbf{v}_2. \tag{29.3}$$

So with $P = [\mathbf{v}_1\ \mathbf{v}_2]$ (invertible, Exercise 29.4), $AP = P\begin{pmatrix} \rho & \omega \\ -\omega & \rho \end{pmatrix}$ and in the coordinates $\mathbf{x} = P\tilde{\mathbf{x}}$:

$$\dot{\tilde{x}} = \rho\tilde{x} + \omega\tilde{y}, \qquad \dot{\tilde{y}} = -\omega\tilde{x} + \rho\tilde{y}. \tag{29.7}$$

Use polar coordinates $\tilde{x} = r\cos\theta$, $\tilde{y} = r\sin\theta$:

1. $r^2 = \tilde{x}^2 + \tilde{y}^2$. Differentiate (chain rule): $2r\dot{r} = 2\tilde{x}\dot{\tilde{x}} + 2\tilde{y}\dot{\tilde{y}} = 2\tilde{x}(\rho\tilde{x} + \omega\tilde{y}) + 2\tilde{y}(-\omega\tilde{x} + \rho\tilde{y}) = 2\rho r^2$, so $\dot{r} = \rho r$.
2. $\theta = \tan^{-1}(\tilde{y}/\tilde{x})$. Chain and quotient rules: $\dot{\theta} = \dfrac{1}{1 + (\tilde{y}/\tilde{x})^2} \cdot \dfrac{\tilde{x}\dot{\tilde{y}} - \tilde{y}\dot{\tilde{x}}}{\tilde{x}^2} = \dfrac{\tilde{x}\dot{\tilde{y}} - \tilde{y}\dot{\tilde{x}}}{r^2} = \dfrac{-\omega\tilde{x}^2 - \omega\tilde{y}^2}{r^2} = -\omega$.
3. Integrate: $r(t) = r(0)e^{\rho t}$, $\theta(t) = \theta(0) - \omega t$.

- $\rho < 0$**: stable spiral** – the radius decays while the angle turns uniformly.
- $\rho > 0$**: unstable spiral.**
- $\rho = 0$**: centre** – closed orbits (ellipses in the original coordinates), periodic with period $2\pi/\omega$; stable but not attracting.

DIRECTION OF ROTATION (ROBINSON SEC. 29.3, EXAMPLE 29.3)

The sign of ω alone does not tell you the sense of rotation in the original plane, because P may reflect. Check a single velocity instead: on the positive y-axis ($x = 0, y > 0$) evaluate $\dot{x}$. If $\dot{x} > 0$ the motion at the top is to the right: **clockwise**. For $\dot{x} = 2x + 5y$, $\dot{y} = -2x$: $\dot{x} = 5y > 0$, clockwise; confirmed on the positive x-axis, where $\dot{y} = -2x < 0$ (downwards).

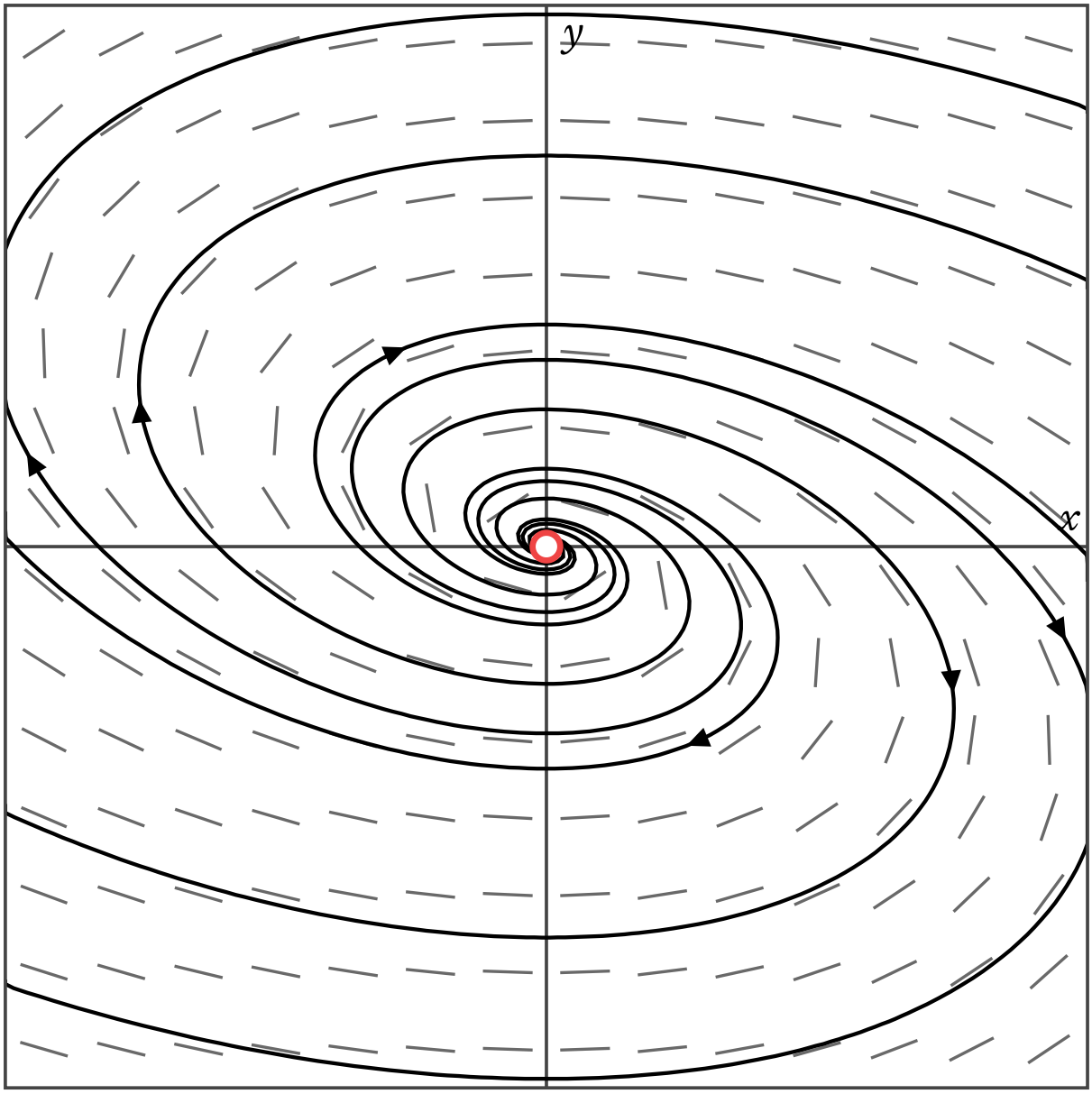


Figure 4.3. *Phase portrait of $\dot{x} = 2x + 5y$, $\dot{y} = -2x$ (Robinson Example 29.3, Fig. 29.6): eigenvalues $1 \pm 3i$, an unstable spiral turning clockwise (on $x = 0$, $y > 0$: $\dot{x} = 5y > 0$).*

8. A repeated eigenvalue: stars and improper nodes

8.1 Stars [Sec. 30.1, p. 295]

If $A = \lambda I$, every vector is an eigenvector and $\mathbf{x}(t) = e^{\lambda t}\mathbf{x}(0)$: solutions move along straight lines through the origin. A **stable star** if $\lambda < 0$, an **unstable star** if $\lambda > 0$.

8.2 Improper nodes [Sec. 30.2, pp. 295–298]

If $A \neq \lambda I$ there is only one eigenvector direction $\mathbf{v}$, so (28.1) cannot be used. Robinson builds a second basis vector:

1. Choose any $\mathbf{v}_2$ not parallel to $\mathbf{v}$ and set $\mathbf{v}_1 = (A - \lambda I)\mathbf{v}_2$. One shows $\mathbf{v}_1$ is an eigenvector (Robinson p. 296; equivalently $(A-\lambda I)^2 = O$ by Cayley–Hamilton, Exercises 30.2–30.3). Hence $A\mathbf{v}_2 = \mathbf{v}_1 + \lambda\mathbf{v}_2$ (30.1).
2. With $P = [\mathbf{v}_1\ \mathbf{v}_2]$: $AP = [\lambda\mathbf{v}_1\ \ \mathbf{v}_1 + \lambda\mathbf{v}_2] = P\begin{pmatrix}\lambda & 1\\ 0 & \lambda\end{pmatrix}$, so $\dot{\tilde{x}} = \lambda\tilde{x} + \tilde{y}$, $\dot{\tilde{y}} = \lambda\tilde{y}$ (30.2)–(30.3).
3. The second equation gives $\tilde{y} = Ae^{\lambda t}$. The first becomes $\dot{\tilde{x}} - \lambda\tilde{x} = Ae^{\lambda t}$; integrating factor $e^{-\lambda t}$: $\frac{d}{dt}\left(e^{-\lambda t}\tilde{x}\right) = A$, so $e^{-\lambda t}\tilde{x} = At + B$.

$$\mathbf{x}(t) = \left[Ate^{\lambda t} + Be^{\lambda t}\right]\mathbf{v}_1 + Ae^{\lambda t}\mathbf{v}_2. \tag{30.4}$$

The factor $te^{\lambda t}$ is the vector version of the repeated-root solution of Topic 3. As $t \to \pm\infty$ the t-term dominates, so trajectories become parallel to $\mathbf{v}_1$ at both ends and bend in an S shape: an **improper node**, stable if $\lambda < 0$ and unstable if $\lambda > 0$.

EXAMPLE 30.1 (P. 298): $A = \begin{pmatrix}5 & -4\\ 1 & 1\end{pmatrix}$

$\lambda^2 - 6\lambda + 9 = 0$, $\lambda = 3$ twice, eigenvector $(2,1)^T$ only. Choosing $\mathbf{v}_2 = (1,0)^T$: $\mathbf{v}_1 = (A - 3I)\mathbf{v}_2 = (2,1)^T$, an eigenvector as claimed. Orientation of the S: on $x = 0, y > 0$, $\dot{x} = -4y < 0$, so trajectories cross the positive y-axis moving left. Unstable improper node.

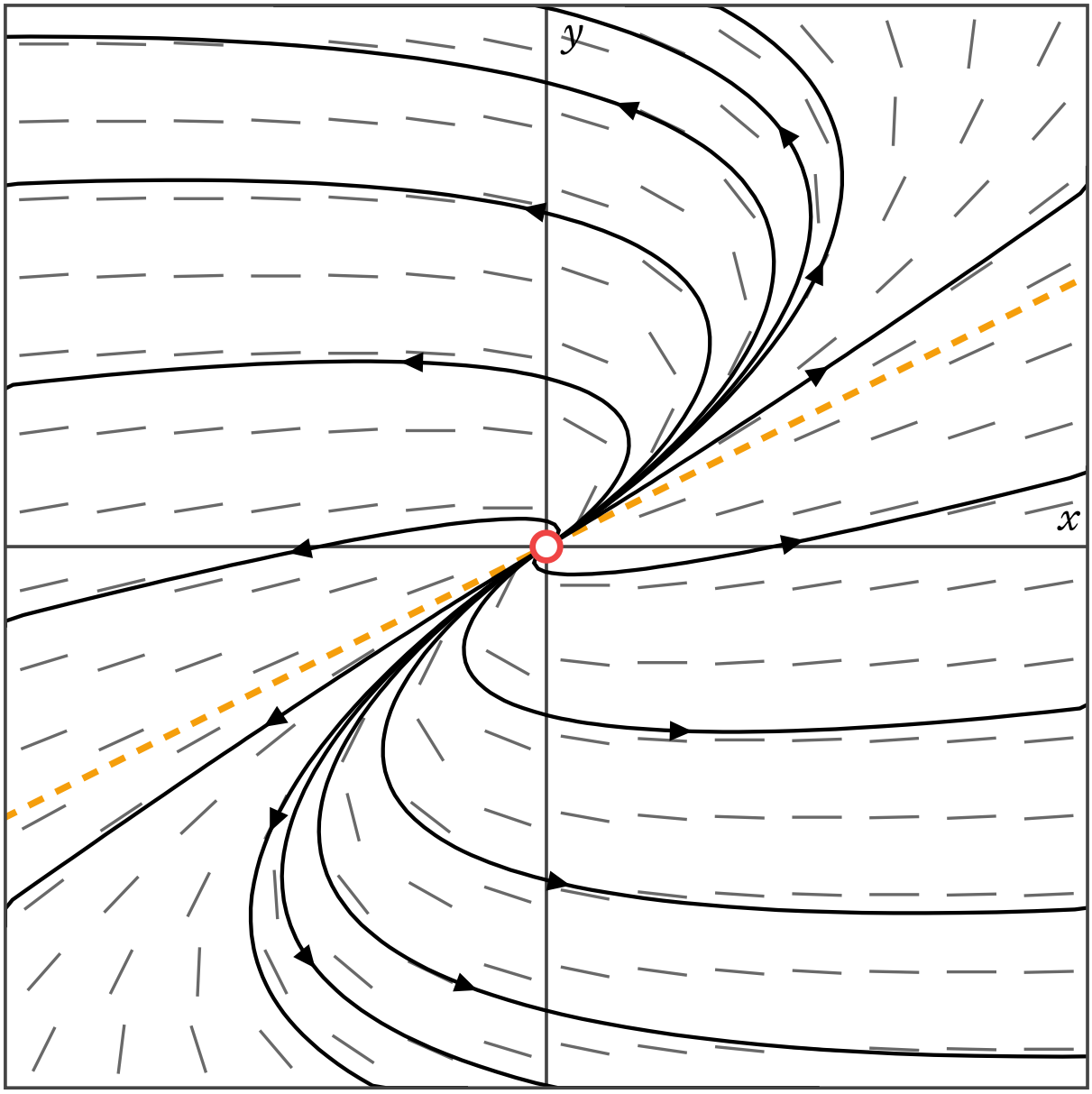


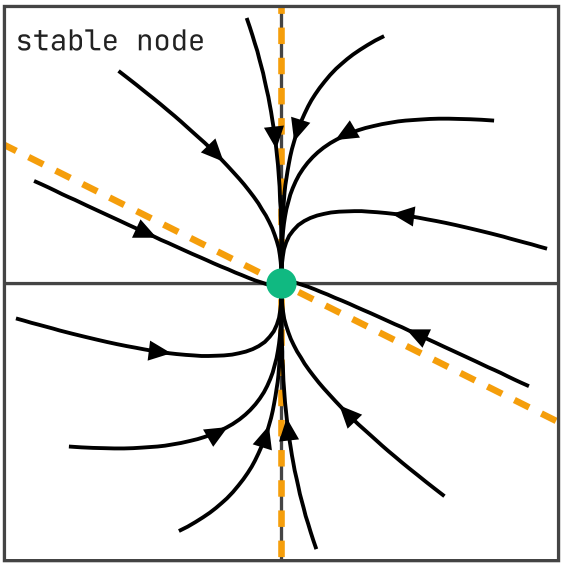
Figure 4.4. *Phase portrait of $\dot{x} = 5x - 4y$, $\dot{y} = x + y$ (Robinson Example 30.1, Fig. 30.3): a repeated eigenvalue $\lambda = 3$ with the single eigenvector $(2, 1)$ (dashed). The origin is an unstable improper node with S-shaped trajectories.*

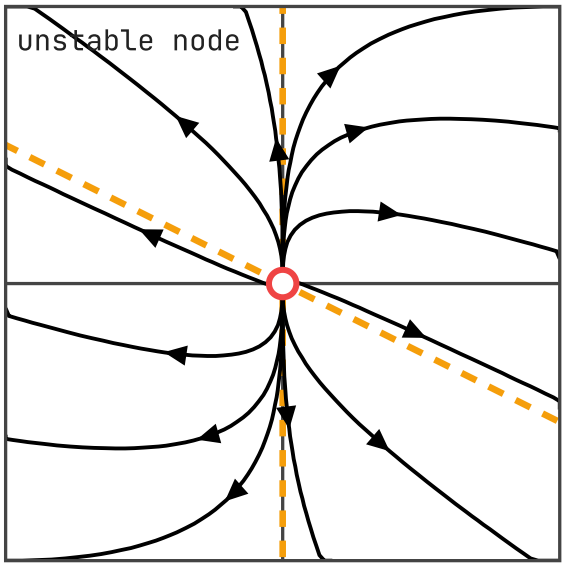
SECTION 09 • ROBINSON CH. 31

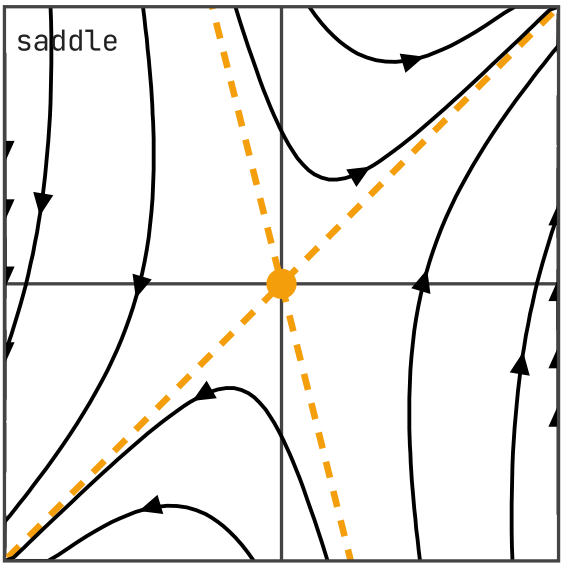
9. Classification of linear phase portraits

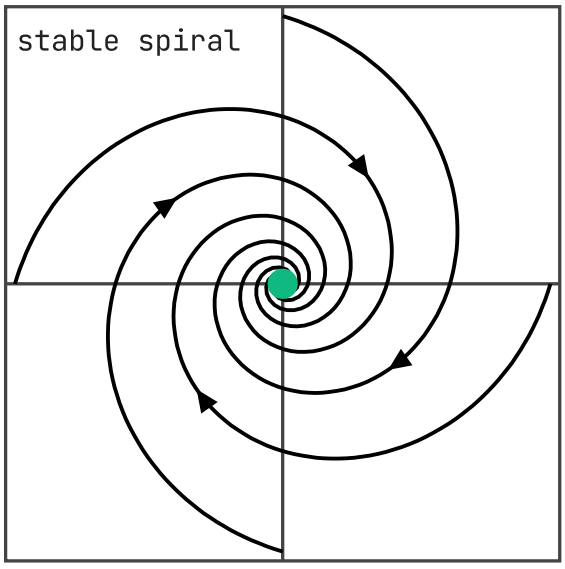
Robinson's summary [Ch. 31, p. 301]: compute the eigenvalues of A; if they are real, compute the eigenvectors.

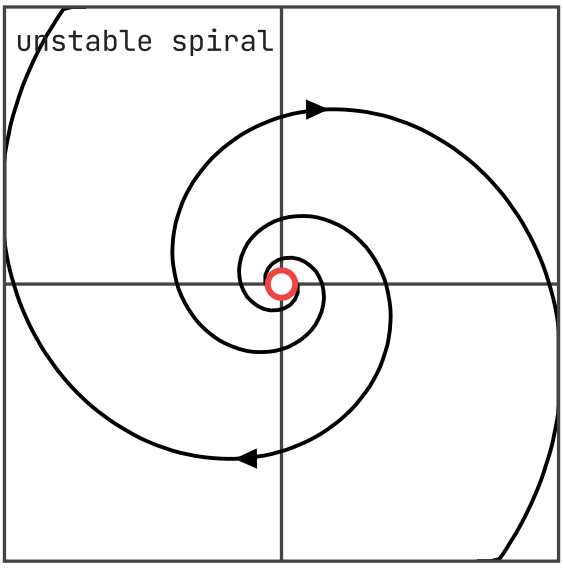
EIGENVALUES	CONDITION	ORIGIN IS A	BEHAVIOUR
Real, distinct	$\lambda_1 < \lambda_2 < 0$	stable node	All trajectories approach $\mathbf{0}$, tangent to the eigenvector of λ_2 (smaller $\lvert\lambda\rvert$)
	$\lambda_1 > \lambda_2 > 0$	unstable node	All trajectories leave $\mathbf{0}$, tangent to the eigenvector of λ_2
	$\lambda_1 < 0 < \lambda_2$	saddle	Only trajectories on the stable eigenvector reach $\mathbf{0}$
Complex $\rho \pm i\omega$	$\rho < 0$	stable spiral	Spiral in; sense from the sign of $\dot{x}$ on $x = 0$
	$\rho > 0$	unstable spiral	Spiral out
	$\rho = 0$	centre	Closed periodic orbits
Repeated λ	$A = \lambda I$	star	Straight lines; stable if $\lambda < 0$
	$A \neq \lambda I$	improper node	S-shaped; stable if $\lambda < 0$

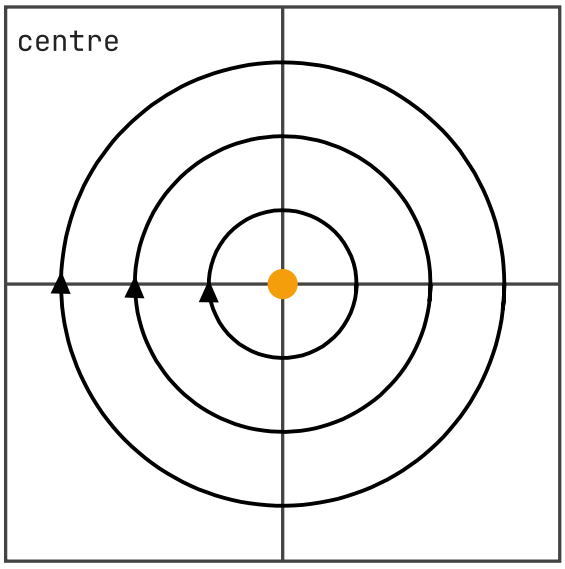

stable node


unstable node


saddle


stable spiral


unstable spiral


centre

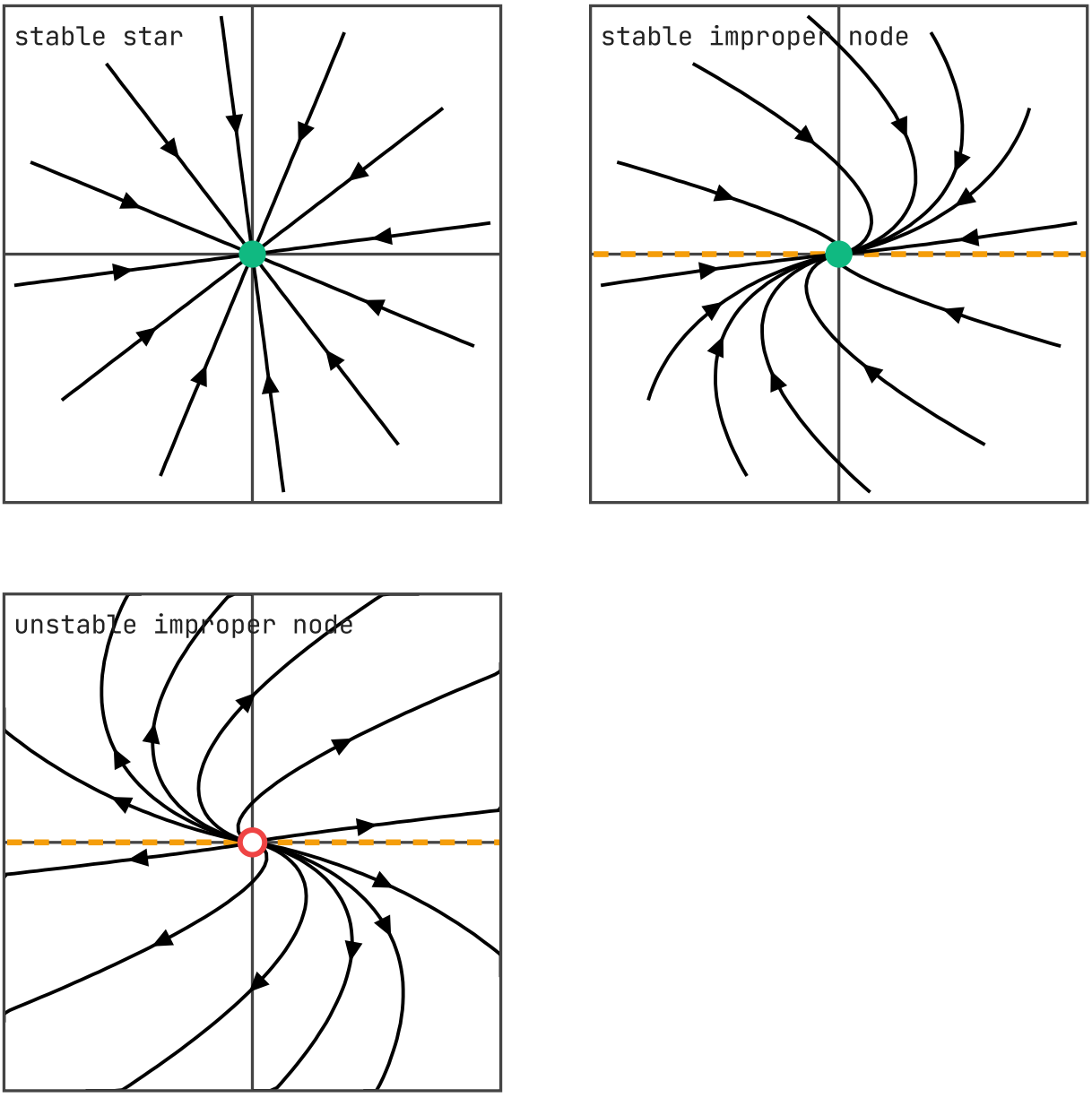


Figure 4.5. *The nine linear phase portraits, arranged as in Robinson Fig. 31.1 (p. 302): first row real distinct eigenvalues, second row complex eigenvalues, third row a repeated eigenvalue. Drawn by numerical integration for representative matrices; dashed amber lines are eigenvector directions.*

9.1 The trace–determinant shortcut (complementary)

Since the eigenvalues solve $\lambda^2 - T\lambda + D = 0$ with $T = \operatorname{tr} A = \lambda_1 + \lambda_2$ and $D = \det A = \lambda_1\lambda_2$, the type can be read off T and D without computing the eigenvalues:

- $D < 0$: real eigenvalues of opposite sign – **saddle**.
- $D > 0$ and $T^2 > 4D$: real, same sign as T – **node** (stable if $T < 0$).
- $D > 0$ and $T^2 < 4D$: complex with $\rho = T/2$ – **spiral** (stable if $T < 0$), or **centre** if $T = 0$.
- $T^2 = 4D$: repeated eigenvalue – star or improper node.

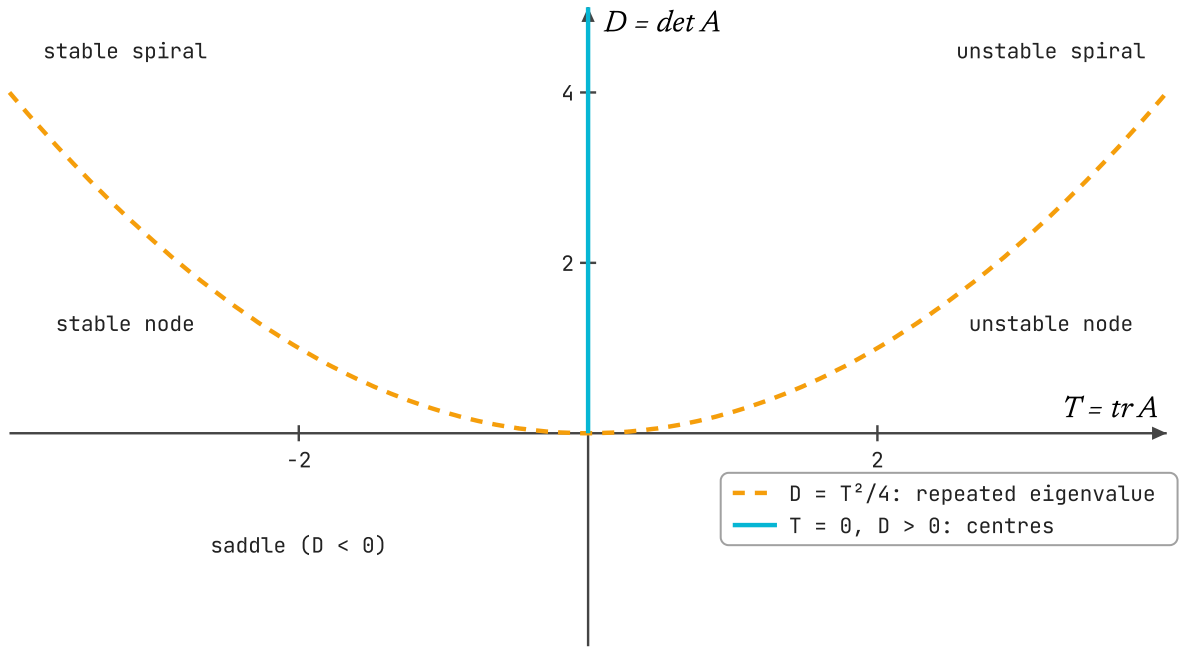


Figure 4.6. *The trace–determinant plane. Since $\lambda^2 - T\lambda + D = 0$, the eigenvalues are real when $T^2 \geq 4D$ (below the dashed parabola, on which they are repeated: stars and improper nodes) and complex above it. This chart is a standard summary that is not in Robinson, who classifies by the eigenvalues directly (Ch. 31).*

THE MASS-SPRING-DAMPER IN THE PHASE PLANE

For $A = \begin{pmatrix} 0 & 1 \\ -\omega^2 & -\lambda \end{pmatrix}$ (Section 1), $T = -\lambda$ and $D = \omega^2 > 0$, and the characteristic equation $\mu^2 + \lambda\mu + \omega^2 = 0$ is the auxiliary equation of Topic 3. The damping regimes are exactly the portrait types:

DAMPING	CONDITION	PHASE PORTRAIT IN $(x, \dot{x})$
none	$\lambda = 0$ ($T = 0$)	centre: energy ellipses $\frac{1}{2}\dot{x}^2 + \frac{1}{2}\omega^2 x^2 = E$
under-damped	$\lambda^2 < 4\omega^2$	stable spiral, clockwise (on $x = 0$, $\dot{x} > 0$ means x increases)
critical	$\lambda^2 = 4\omega^2$	stable improper node ($A \neq \lambda I$ always, since the top-left entry is 0)
over-damped	$\lambda^2 > 4\omega^2$	stable node

The same classification underlies flight dynamics: linearising the longitudinal equations of an aircraft about trimmed flight gives a matrix whose complex eigenvalue pairs are the short-period and phugoid modes, and their real parts decide whether the aircraft returns to trim after a gust.

LECTURE COVERAGE

By Oct 8 the lecture had covered Sections 1–6 (Ch. 25–28 Sec. 28.4). Sec. 28.5 (stable and unstable manifolds), Ch. 29 (complex eigenvalues), Ch. 30 (repeated eigenvalue) and Ch. 31 (summary) were still ahead; they are included here so that the topic is complete.